

Importance Control - Gaze-Based Adaptation Objective

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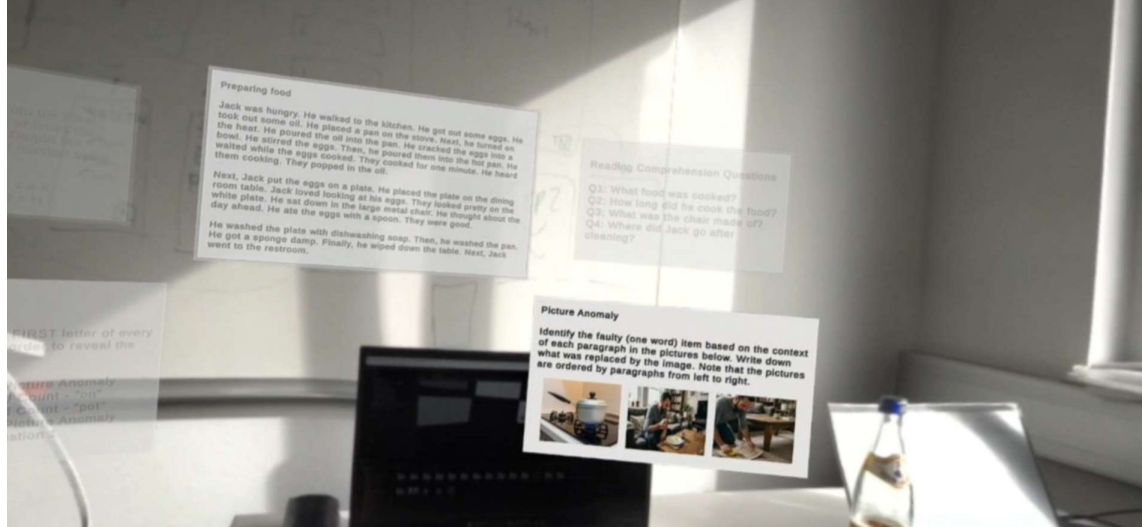


Fig. 1. Puzzle tasks using Importance Control

Current XR adaptation toolkits mainly support single-window optimization and provide limited support for multi-window interfaces, where visual clutter can increase cognitive burden. To address this, we present Importance Control, a gaze-based adaptation objective for the Adaptive User Interfaces Toolkit (AUIT). The approach estimates the relevance of UI windows from gaze behaviour and adapts opacity and scale to reduce the prominence of less relevant elements. We implemented the method in Unity using Meta Quest Pro eye tracking and evaluated it in a within-subjects user study with seven participants. Participants completed two reading-comprehension tasks in a mixed reality environment, once with Importance Control and once without it. Results indicate that the system was generally rated positively, especially regarding responsiveness, robustness, and overall preference. Participants also reported reduced perceived cognitive burden. However, task-completion times did not show a clear performance advantage, mainly due to strong learning effects across tasks. Overall, the findings suggest that gaze is a promising context source for adaptive multi-window XR interfaces

1 INTRODUCTION

The AUIT as well as other contemporary UI Adaptation Toolkits, mainly tackle objectives adapting single windows/applications, but when confronted with multiple interfaces, the vision becomes cluttered, obscuring both the physical world and the virtual content leading to an increase in cognitive load, forcing the users to manually reshape and organize the UI elements. To close this gap, we propose “Importance Control”, an extension of adaptation objectives to the AUIT that supports developers designing complex, multi-window applications. By assessing user attention on high-priority windows using Importance Control, visual clutter is reduced autonomously by adapting the visibility of less prevalent windows, which we achieve by featuring gaze-based context source by measuring three variables:

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frequency, recency, and dwell time. We claim that Importance Control can be utilized in the development of adaptation methods to enhance users' performance and satisfaction. To support our hypothesis, we will conduct a user study.

2 RELATED WORK

The idea of adapting numerous windows autonomously and simultaneously to reduce visual clutter was inspired by the papers that focused on the elimination of real-world object appearances in order to minimize distractions in a workspace [Yokoro et al. 2023] and the studies of visual noise that interfere with the productivity and comfort in a workshop environment [Hong et al. 2024]. We made the connection that since there is correlation between perceptual information overload and mental as well as physical strain, it naturally applies to virtual optical obstructions caused by the accumulation of UI elements. The concept of adaptation objectives proposed by the AUIT paper [Evangelista Belo et al. 2022] instigated the proposition of an additional adaptation objective which we call "Importance Control" – stemming from the notion of learned user preference in adaptation objectives [Song et al. 2025] – to bridge the visual salience optimization in a multi-window setting. We made the decision to replace user preference by sampling gaze as our main context source which can be realized by EyeMRTK [Mardanbegi and Pfeiffer 2019], a toolkit for developing eye gaze interactive applications. Using the toolkit, we intend to acquire attentional metrics through gaze-based intent inference i.e. dwell time [Pfeuffer et al. 2021]. To summarize, we came up with Importance Control, an adaptation objective that utilizes gaze information.

3 METHOD

Our methodology is divided into four main components that together form the adaptive user interface pipeline. First, we describe the acquisition and processing of gaze data. Second, we introduce the importance computation, a model that estimates the relevance of each UI window based on user gaze behavior, which serves as the primary context source. Third, we define the Importance Control adaptation objective, which integrates the computed importance into the Adaptive User Interface Toolkit (AUIT) by means of a cost function and an optimization rule. Finally, we present the transition property, responsible for applying UI adaptations smoothly over time to ensure a consistent user experience.

3.1 Gaze Data

This section describes how gaze information is acquired, processed, and stored for further use in adaptive user interface optimization.

3.1.1 Goal. The goal of this component is to collect reliable gaze-based interaction data that can be used to infer user preferences. This data serves as a foundation for downstream computations, particularly for estimating the relevance of individual interface elements.

3.1.2 Challenges. Several challenges arise when acquiring gaze data in an XR environment:

- Setting up accurate and stable eye tracking within Unity using the Meta Quest Pro hardware.
- Detecting meaningful gaze interaction events such as gaze enter and gaze exit.
- Associating gaze events with specific UI elements.
- Ensuring low computational overhead while maintaining responsiveness.
- Handling noisy gaze signals and unintended short glances.

3.1.3 Solution. We leverage the Unity Asset Store package [<https://assetstore.unity.com/packages/tools/camera/vr-gaze-interaction-system-241337>] in combination with the Meta XR SDK to handle low-level gaze tracking. The system uses a Gaze Interactor attached to the eye tracking object (in our case the left eye anchor) and Gaze Interactable components attached to UI elements. These components emit interaction events that abstract away raw gaze data into semantically meaningful signals. This approach allows us to work at the level of interaction events instead of raw eye vectors, simplifying both implementation and reasoning.

3.1.4 Implementation. A Gaze Interactor prefab is attached to the EyeGazeLeft object within the XR rig. Each UI window is augmented with a Gaze Interactable component. The system triggers the following events:

- Gaze Enter → user starts looking at a window
- Gaze Exit → user stops looking at a window

To mitigate the "Midas Touch" problem and filter out ambiguous focus (such as short glances or accidental passes), we set the activation threshold at 0.3 seconds inside the Gaze Interactor. This activation rate ensures that only intentional gazes are captured and stored within the WindowGazeData class. This class maintains various other parameters and variables we will discuss in the Importance Formula section.

3.2 Importance Formula

This section defines the core context model used to quantify the relevance of UI elements.

3.2.1 Goal. The goal is to compute an importance value for each UI window that reflects its relevance to the user over time. This importance measure is used to:

- Infer user interest
- Guide UI adaptation
- Reduce visual clutter by prioritizing relevant elements

The underlying hypothesis is that user gaze behavior correlates with perceived importance, meaning that frequently and recently observed elements should be considered more relevant.

3.2.2 Challenges. Designing the importance function involves several constraints:

- Computational efficiency (must run in real-time)
- Avoiding long-term memory bias (old interactions dominating)
- Avoiding short-term bias (only recent interactions dominating)
- Managing memory usage
- Preserving correlations between metrics (frequency, dwell time, recency)
- Ensuring interpretability for tuning and debugging
- Selecting an appropriate update frequency

3.2.3 Solution. We adopt a time-decayed accumulation model that integrates three key metrics:

- Frequency of gaze interactions
- Dwell time (continuous attention)
- Recency via exponential decay

The importance is updated incrementally using exponential decay:

$$F_{\text{new}}(t) = F_{\text{old}} * e^{-\lambda_{\text{Freq}}(t-t_{\text{last}})} + \text{Indicator}(\text{GazeEnter}) \quad (1)$$

$$I_{\text{new}}(t) = I_{\text{old}} * e^{-\lambda_{\text{Imp}}\Delta t} + a(F_{\text{new}} * W_F + \text{Indicator}(\text{GazeStay}) * W_D) \quad (2)$$

Where:

- λ controls decay rate (note: each formula uses its own respective decay rate)
- W_F weights frequency contribution
- W_D rewards sustained attention
- Indicator functions capture discrete gaze events

This formulation:

- Prevents unbounded growth (via decay)
- Balances short-term and long-term effects
- Maintains low memory usage
- Preserves metric relationships

3.2.4 Implementation. Each UI window maintains an instance of WindowGazeData with the prescribed update functions above. An Update Manager regulates the calculation frequency for both importance and frequency metrics, currently operating at a fixed polling interval of 0.5 seconds. Furthermore, it serves as a global configuration provider, exposing all necessary hyperparameters to the developer. To maintain systemic consistency and prevent exceptions, the Update Manager is the exclusive interface for parameter modification; external adjustments are strictly discouraged in the current implementation.

3.2.5 Parameter Selection. To ensure system stability, we derived the theoretical upper bound of the importance value. Under conditions of overdue frequency ($F_{\text{new}} = 0$), indefinite sustained attention ($\text{Indicator}(\text{GazeStay}) = 1$) and a fixed update interval Δt , the importance I converges to a steady-state value I_{∞} :

$$I_{\infty}(t) = \frac{a * W_D}{1 - e^{-\lambda_{\text{Imp}} * \Delta t}} \quad (3)$$

The weight and decay of the frequency can then be determined independently.

3.3 Importance Control Adaptation Objective

This section describes how the computed importance values are integrated into the Adaptive User Interface Toolkit (AUIT) to drive UI adaptation.

3.3.1 Goal. The goal is to translate the importance values of UI windows into a form that can be used by AUIT to guide adaptive behavior. This includes computing a normalized cost value for each window, generating optimization suggestions that reflect their relative importance, and enabling dynamic adaptation of UI properties such as opacity and scale.

3.3.2 Challenges. Several challenges arise in this process:

- Importance values are unbounded, while AUIT requires a cost value in the range [0,1].
- Adaptation should respond immediately to user gaze without waiting for importance to accumulate.

- UI changes must remain interpretable and consistent with user expectations.

3.3.3 Solution. To address these challenges, we map importance values to a bounded function using a sigmoid-based normalization:

$$I_{norm} = \frac{1}{1 + e^{-gI} * (\frac{1}{l} - 1)} \quad (4)$$

Where:

- g controls the growth speed
- I is the importance
- l creates a lower bound

This ensures that importance values are constrained to the interval $[l, 1]$ with $0 < l < 1$. These values are then directly used in the cost function as a measure of opacity comparison. However, to account for the momentary persistent focus ($GazeStay = true$), we must formalize a piecewise definition that encapsulates the transition between historical importance and real-time interaction events. Thus the cost function is defined as:

$$Cost = \min(\alpha_{cost}, G_{cost}) \quad (5)$$

Where:

$$\alpha_{cost} = |I_{norm} - \alpha_{curr}| \quad (6)$$

$$G_{cost} = \begin{cases} 1 - \alpha_{curr} & , \text{ if } GazeStay = true \\ 1 & \text{ otherwise} \end{cases} \quad (7)$$

In addition, we define an optimization rule that generates candidate UI configurations by adjusting visual properties based on importance. Specifically, windows with higher importance are assigned higher opacity and larger scale, while less important windows are reduced in prominence. For this, the generated values are described by:

$$\alpha_{new} = I_{norm} \quad (8)$$

$$\beta_{new} = (I_{norm})^r \quad (9)$$

Where α_{new} represents the new suggested opacity and β_{new} represents the new size/scale. r is the resize coefficient used to control the effect while keeping the values proportional to the resulting opacity.

This establishes a monotonic relationship between importance and visual emphasis, ensuring that adaptation aligns with inferred user attention.

3.3.4 Implementation. We implement a custom AUIT objective called:

ImportanceControl

This objective consists of two core components:

Cost Function. The cost function computes a normalized cost value from the current importance of each window using the sigmoid mapping described above. This value is used by AUIT to evaluate the quality of different UI configurations.

Optimization Rule. The optimization rule generates new candidate configurations by adjusting UI parameters such as opacity (α) and scale (β) according to window importance. Specifically:

- $\alpha \propto I$ (higher importance $\rightarrow$ higher opacity)

- $\beta \propto I$ (higher importance $\rightarrow$ larger scale)

These mappings are applied within predefined bounds to ensure visual stability. The generated configurations are then evaluated using the cost function, and AUIT selects the configuration that minimizes the overall cost across all windows.

3.3.5 Parameter Selection. By defining a lower bound β_{min} for the UI scale, the resizing factor r can be calculated, such that the window scale converges to the selected β_{min} as Importance I approaches l , using the following formula:

$$r = \frac{\log(\beta_{min})}{\log(l)} \quad (10)$$

The resizing coefficient r can be configured in the *UpdateManager*.

3.4 Transition Property

This section describes how UI adaptations are applied over time to ensure smooth and visually consistent transitions.

3.4.1 Goal. The goal of the transition property is to apply changes to UI elements in a smooth and gradual manner, avoiding abrupt visual updates that could disrupt the user experience. This is particularly important in XR environments, where sudden changes can reduce usability and increase cognitive load.

3.4.2 Challenges. Applying adaptive changes in real-time introduces several challenges:

- Directly applying new parameter values can result in abrupt and visually jarring transitions.
- Frequent updates from the optimization process may lead to overlapping or conflicting transitions.
- Ensuring temporal consistency while maintaining responsiveness to user input.
- Balancing smoothness with adaptation speed.

3.4.3 Solution. We introduce a time-based interpolation mechanism that incrementally updates UI properties over a fixed transition interval. Instead of immediately applying new values, the system gradually interpolates between the current state and the target state.

To prevent conflicting updates, a locking mechanism ensures that only one transition is active at a time. New adaptation requests are ignored while a transition is ongoing, preserving consistency.

3.4.4 Implementation. The transition system operates on key UI parameters:

- Opacity (α)
- Scale (β)

When a new configuration is selected by AUIT, the system:

1. Stores the target values for each parameter
2. Gradually updates the current values toward the target values over a fixed duration (currently 0.5 seconds)
3. Applies updates incrementally on each frame

A boolean flag is used to indicate whether a transition is currently active. If true, additional adaptation updates are temporarily blocked to avoid overlapping transitions.

This approach ensures smooth, continuous changes while maintaining a stable and predictable user interface.



Fig. 2. Smooth transitions using time-based interpolation mechanism

4 USER STUDY

To evaluate the proposed adaptation objective, we conducted a summative user study in an MR environment. The study aims to investigate whether the proposed Importance Control affects perceived workload, user experience, and user preference during task execution. Furthermore, the study explored whether gaze-based adaptation can support users in an AR setting with multiple simultaneously displayed interface elements. The study is based on the following hypotheses:

H1. *Gaze serves as an effective method for user preference inference.*

H2. *Importance Control provides a satisfying user experience.*

H3. *Importance Control reduces cognitive load.*

4.1 Participants

A total of seven participants took part in the study (2 female, 5 male, age: $M = 23.29$, $SD = 2.29$). No prior programming experience or previous experience with augmented reality was required. Since the study tasks involved reading-comprehension exercises in English, participants were required to have at least B1-level English proficiency. In addition, participants needed sufficient cognitive and physical ability to read and write during the experiment. Participants with visual impairments, such as wearing glasses, were not excluded.

4.2 Study Design

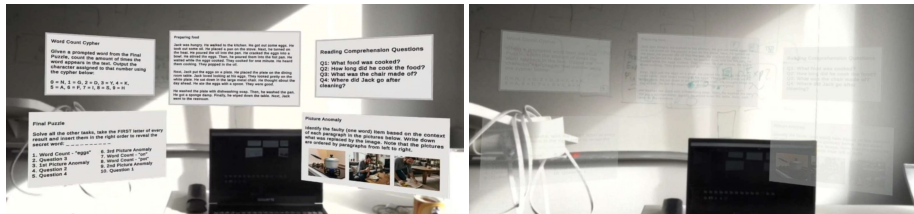


Fig. 3. Left: Puzzle without Importance Control, Right: Puzzle with Importance Control

The study followed a within-subjects design in which each participant completed tasks under both experimental conditions: with Importance Control enabled and without Importance Control. To reduce bias caused by increasing familiarity with the task and the system, the order of conditions was counterbalanced. Three participants completed the first task with Importance Control enabled and the second task without, whereas the remaining four participants completed the tasks in the opposite order.

4.3 Procedure

At the beginning of the study, demographic and background information was collected, including age, gender, prior AR/VR experience, English proficiency, and visual impairments. In Addition, participants were asked to fill in a consent form. Participants were then instructed to put on the Meta Quest Pro headset and were guided through the eye-tracking calibration process. After calibration, they entered a short introductory environment containing a small number of windows intended to familiarize them with the AR setup and to verify that the calibration had been successful. This introductory phase did not include the Importance Control mechanism. Following this familiarization phase, participants received an introduction to the main task. They were informed that they would complete two B1-level reading-comprehension task in AR and that they could use a tablet to take notes and write down their answers. The tasks were comprised of five windows – a central window displaying the reading text with the surrounding windows containing task-related content. Participants then completed both tasks, one under each experimental condition, with the order depending on the assigned counterbalancing group. After completing both tasks, participants filled in a post-study questionnaire including a feedback section.

4.4 Measures

The study collected both objective and subjective measures. As an objective measure, task completion time was recorded. Subjective measures were obtained using a custom questionnaire based on a 5-point Likert scale. The questionnaire included items on perceived task difficulty, mental and physical demand, system responsiveness and consistency, perceived necessity and satisfaction. For further details, see the questionnaire in [Appendix A](#).

5 RESULTS

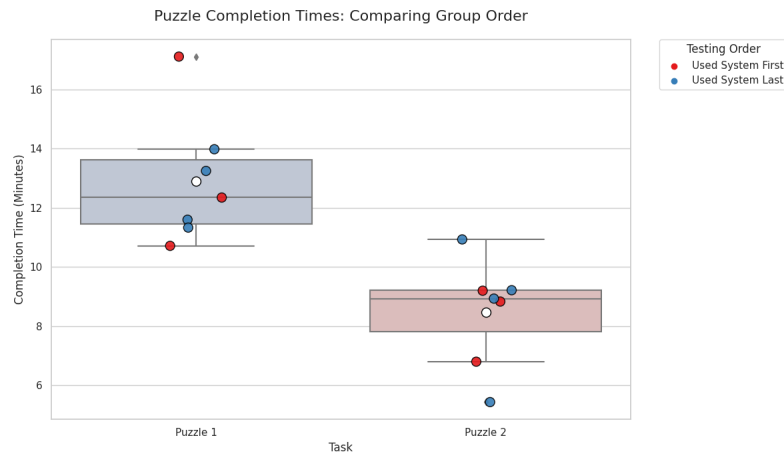


Fig. 4. Box plot depicting average puzzle completion time with deviations

Given the small sample size ($n=7$), the findings are interpreted primarily on a descriptive level. The analysis combines objective task-completion times with subjective questionnaire responses. We grouped the statements of the questionnaire into 4 different categories for easier interpretation: Cognitive Load without IC (1, 11), Cognitive Benefits from IC (2, 3, 12), Responsiveness and Robustness (4, 5, 6), and User Preference (7, 8, 9, 10).

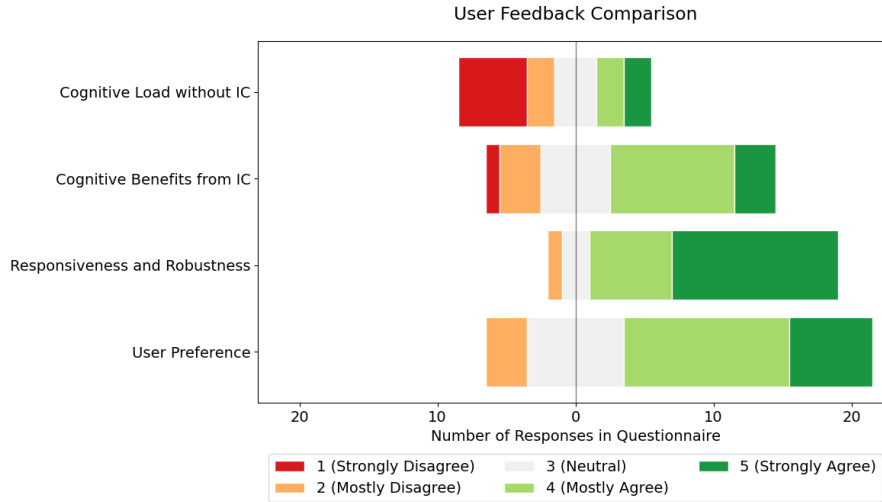


Fig. 5. Diverging stacked bar chart including all user ratings
 We measure the total user ratings over 4 subcategories of statements in the questionnaire
 (Note: Cognitive Load without IC ratings measure the level of cognitive load on the puzzle and environment in general)

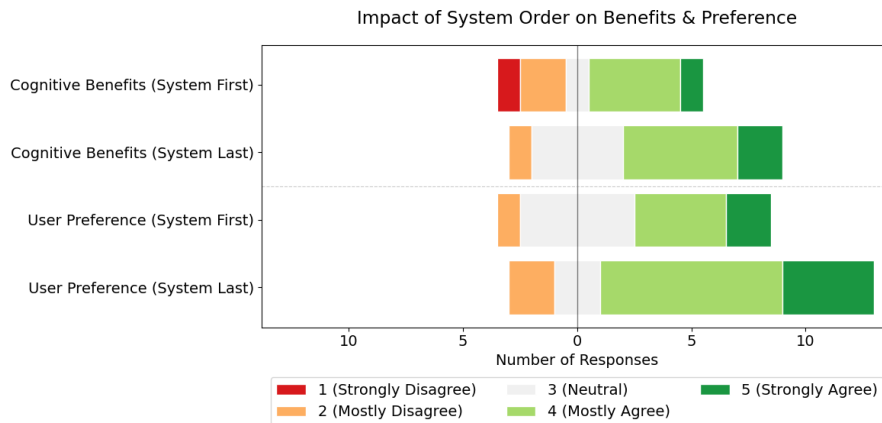


Fig. 6. Diverging stacked bar chart accounting for system order bias of the two groups

5.1 H1: Gaze serves as an effective method for user preference inference

Participants generally rated the adaptive system positively, particularly in the categories Cognitive Benefits from IC and User Preference, suggesting that the system's behavior was perceived as helpful and aligned with users' needs. The positive tendency was even stronger when participants experienced the adaptive system after first using the non-adaptive condition, indicating that the adaptive behavior was noticeable and, in many cases, preferred. The study provides evidence that gaze can function as a useful context source for preference-related adaptation.

However, this hypothesis is only partially supported, not conclusively proven. The study shows that users responded positively to the gaze-driven adaptation, but it does not directly measure how accurately gaze inferred explicit user preferences. No ground-truth comparison between inferred preference and self-reported preference was conducted. Therefore, the results suggest that gaze is a promising and practically useful method for user preference inference, but stronger validation would require a more direct evaluation of inference accuracy.

5.2 H2: Importance Control provides a satisfying user experience

Participants largely agreed that the system was responsive and well integrated, and the overall preference-related items also showed a predominantly positive distribution. The strongest positive responses in the questionnaire were observed in the categories of Responsiveness and Robustness and User Preference.

These results indicate that the adaptive system was not only technically acceptable but also subjectively well received. In particular, the strong agreement in the preference-related items suggests that users generally perceived the system as beneficial rather than unnecessary or distracting.

At the same time, the order-specific results indicate that participants who experienced the adaptive system last rated it more positively than those who experienced it first. This suggests that part of the positive evaluation may have been amplified by direct comparison with the non-adaptive condition. Nevertheless, even with this possible contrast effect, the overall pattern of responses remains positive. Thus, the results support the conclusion that Importance Control provided a satisfying user experience within the context of this study.

5.3 H3: Importance Control reduces cognitive load

Figure 5 indicates that participants generally did not perceive the puzzles themselves as excessively difficult or visually overwhelming. At the same time, the responses still suggest that Importance Control was associated with perceived cognitive benefits, as participants tended to evaluate the adaptive condition positively on items related to reduced mental demand and reduced interface burden. As shown in Figure 6, these positive ratings were more pronounced among participants who experienced the adaptive system after first using the non-adaptive condition. This pattern may reflect an order-related bias, as the direct comparison with the baseline condition could have made the benefits of Importance Control more salient.

In contrast, the task-completion times do not provide clear evidence that IC reduced cognitive load in a performance-based sense. Figure 4 shows that Puzzle 1 had a mean completion time of 12.91 minutes, whereas Puzzle 2 had a mean completion time of 8.48 minutes, meaning that the second task was completed on average 4.43 minutes faster than the first. This pattern occurred in both order groups. Participants who used the adaptive system first improved from 13.39 to 8.28 minutes, while participants who used it last improved from 12.54 to 8.63 minutes.

This consistent decrease in time for the second puzzle strongly suggests a learning or familiarization effect. When the data are regrouped by condition, the difference between conditions is negligible: the mean completion time was 10.67 minutes with IC and 10.71 minutes without IC. Accordingly, the timing data do not show a meaningful performance advantage for IC. Instead, they indicate that order effects were stronger than any condition effect in the objective measure.

Taken together, these results suggest that Importance Control reduced perceived cognitive load, but this reduction was reflected more clearly in subjective experience than in objective task performance. Therefore, H3 is supported at the level of user perception, but not strongly supported by completion-time data.

6 CONCLUSION

Overall, the findings suggest that the proposed adaptation approach was received positively by participants. The subjective responses indicate that the system was experienced as responsive, beneficial, and generally preferable to the non-adaptive condition. This supports the claim that Importance Control provides a satisfying user experience. The results also suggest that the gaze-based adaptation mechanism was meaningful to users, which offers initial evidence that gaze can serve as an effective method for inferring user-relevant preferences. Finally, the questionnaire results indicate that IC reduced the perceived burden of the fixed multi-window interface, although this benefit was not mirrored by a clear improvement in completion times.

7 LIMITATIONS AND FUTURE WORK

A limitation of the current implementation is that the proposed Importance Control is still restricted in several ways. At present, the adaptation is primarily applied to UI canvas elements and does not yet generalize to all game objects in the scene. In addition, the current approach focuses on a limited adaptation dimension and has not yet been extended to other forms of adaptation, such as repositioning. Another practical limitation is that the update interval or refresh rate is currently hard-coded, which reduces flexibility for developers who may wish to tune the system to different application contexts. Future work should therefore aim to generalize the adaptation objective to additional adaptation types, make Importance Control available across a broader range of scene objects, and expose key hyperparameters such as the refresh interval to programmers. Furthermore, allowing users to manually adjust importance values according to their own preferences could improve personalization and provide a useful complement to the gaze-driven inference mechanism.

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A APPENDIX

The following questionnaire was used to evaluate participants' subjective experience with the system. All closed questions were rated on a 5-point Likert scale:

Questionnaire Items

- (1) I found the tasks hard to solve.

- 573 (2) Using the system, I found the tasks to be less mentally demanding.
574 (3) Using the system, I found the tasks to be less physically demanding.
575 (4) I found the system very responsive.
576 (5) I thought there was too much inconsistency in this system.
577 (6) I found the various functions in this system were well integrated.
578 (7) I prefer using this system.
579 (8) I found the system unnecessary.
580 (9) I found the system very satisfying.
581 (10) I found the system very distracting.
582 (11) The amount of windows was overwhelming.
583 (12) Using the system, the amount of windows was less overwhelming.
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587 **Feedback:**

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589 Please describe any additional impressions, difficulties, or suggestions regarding the system.
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